

**STRULE AUTOMATION**

Industrial Controls · Bay Area & Northern California

PLANT CHECKLIST

MRF Controls Walk-Down Checklist

A one-page checklist to carry through a single- or dual-stream material recovery facility (MRF) when the line keeps stopping, sort quality is slipping, or a retrofit didn't settle in. Work through it with the line running where it's safe to, and locked out where it isn't. It pairs with the full field guide on MRF control systems.

Before you start

- ☐ Confirm you have a current, accurate one-line and sequence drawing (assume it has drifted from the code until proven otherwise)
- ☐ Confirm lockout/tagout (LOTO) is available and you'll use it before reaching into any equipment
- ☐ Have a way to capture time-stamped fault data from the PLC and from major machines (sorters, baler, drives)

Start / stop sequencing and interlocks

- ☐ Belts start from the discharge end back toward the infeed (material always lands on a moving belt)
- ☐ Belts stop from the infeed back toward discharge (each belt clears before the one feeding it stops)
- ☐ Each conveyor's start permit actually reflects the downstream belt running
- ☐ Screens and rotating equipment are up to speed before the conveyor feeding them can run
- ☐ A stalled or tripped belt stops everything feeding it (no material piling onto a dead or buried belt)
- ☐ Baler infeed is interlocked to chamber-full so it stops loading instead of starving or jamming
- ☐ Mapped the interlock chain as it exists in the code, not just as drawn

New / retrofitted equipment (sorters, eddy-current, magnets, screens)

- ☐ Every added machine is in the start/stop sequence and interlock chain like everything else
- ☐ A fault on each machine has a deliberate, documented effect (whole line / its zone / nothing)
- ☐ Optical and near-infrared (NIR) sorters read a clean belt encoder/tach signal (eject timing is slaved to it, so eject position tracks belt speed automatically — check the feedback, not the VFD setpoint)

- ☐ Belt speed matches what the sorter was commissioned with; if speed changed, the sorter was re-taught
- ☐ Trigger photo-eye and encoder registration checked (clean, aimed, coupled) when sort quality has drifted with nothing faulted
- ☐ Material presents to the sorter as a near-monolayer (roughly one item deep), not stacked or shingled where the sensor cannot see individual pieces
- ☐ Sorter stops firing air when its feed belt stops; feed belt stops loading when the sorter faults
- ☐ Checked the "seam" between vendors first when a retrofitted zone misbehaves

Compressed air feeding pneumatic ejectors

- ☐ Measured header pressure at the ejector manifold under full load, not just at the compressor gauge, using a fast/peak-hold gauge or a logger sampling at least a few Hz (a needle gauge damps out the sub-second dip and reads "fine")
- ☐ Watched header pressure during heavy, dense flow (does it sag when many ejectors fire at once?)
- ☐ Under peak ejection, manifold pressure stays above the machine's stated minimum — check the spec plate; supply is commonly ~85–100 psi and the not-below floor often near 70–80 psi
- ☐ Burst droop stays within a few psi; more than ~10–15 psi, or any dip that crosses the sorter's low-pressure trip, means the supply is undersized or restricted
- ☐ Walked the header and fittings for leaks; checked for a stuck or leaking ejector valve (bleeds the header continuously and drives sag even with no target present)
- ☐ Receiver tank capacity is sized for the current ejector count, not the original design; automatic drain is clearing condensate
- ☐ Compressors aren't short-cycling on too tight a pressure band
- ☐ Low header pressure is alarmed so air starvation announces itself instead of hiding as bad sort quality

Throughput and utilization

- ☐ Separated the two questions: is the line slow when running, or does it stop too often?
- ☐ First-out fault is easy to find (which machine tripped first, versus the cascade after it)
- ☐ PLC fault-history/sequence-of-events (SOE) buffer is enabled, so you can read a cascade backward to the trigger
- ☐ Alarm log is clean enough to actually read (noise isn't burying the alarm that matters)
- ☐ Feed into screens and sorters is steady, not slugging in bursts that overload then starve
- ☐ Confirmed the real constraint before turning up belt speed

The multi-vendor seam

- ☐ Identified where two vendors' machines hand off to each other (network, hardwired signals, shared I/O)
- ☐ For any once-an-hour intermittent fault, logging is in place on both sides of the handshake
- ☐ Checked shared networks for drops under load, and grounding around large drives
- ☐ No single vendor is being asked to own a problem that lives between two of them

Safety

- ☐ Every e-stop drops the zone a person can actually reach from it
- ☐ No e-stop has been quietly disconnected or rezoned by a past retrofit
- ☐ Guard interlocks stop the right equipment when a guard opens, and none are jumpered out
- ☐ LOTO is real and enforced at every routine jam-clearing point, including bleeding stored pneumatic energy before anyone reaches into a sorter
- ☐ Any nuisance trip is scheduled to be fixed at its cause rather than defeated or jumpered

If the walk-down turns up something that won't settle — especially a fault that lives in the handoff between two machines, where no single vendor will own it — that's the kind of thing I get called in for. Reach me at struleautomation.com or info@struleautomation.com. I work across the Bay Area and Northern California.